JIARUI WANG

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Summary

My research focuses on memory architectures for large language models, aiming to decouple knowledge storage from reasoning capabilities and enable models to continually acquire, retain, and update knowledge. To this end, I develop scalable, modular parametric memories that can be trained independently and integrated with modern LLM backbones. Recently, I led the research and development of *Intern-S2-Preview*'s memory architecture for continual domain expansion. Beyond memory architectures, I also work on LLM pretraining, supervised fine-tuning, data pipelines, and inference systems.

Education

Shanghai Jiao Tong University, Shanghai, China — Sept. 2025 – Jun. 2030
Ph.D. in Computer Science and Technology
Advisor: Prof. **Zhouhan Lin** (LUMIA Group)

Shanghai Jiao Tong University, Shanghai, China — Sept. 2021 – Jun. 2025
B.S. in Computer Science and Technology

Publications

- **MemSFT: Mitigating Alignment Tax with an External Parametric Memory**
  Jiarui Wang*, Xiang Shi*, Jiaqi Cao, Rubin Wei, Xiquan Wang, Hao Sun, Jingzhi Wang, Zhiqi Yang, Qipeng Guo, Bowen Zhou, Zhouhan Lin
  arXiv:2607.25614, 2026.
- **Intern-S2-Preview: Scientific Agentic Foundation Model**
  Intern-S2-Preview Team, Shanghai AI Laboratory; Co-author; *arXiv:2608.13505, 2026.*
- **Memory Decoder at Scale: A Pretrained, Parametric Long-Term Memory**
  Rubin Wei, Jiaqi Cao, **Jiarui Wang**, Junming Zhang, Qipeng Guo, Bowen Zhou, Zhouhan Lin
  arXiv:2607.27919, 2026.
- **MLP Memory: A Retriever-Pretrained Memory for Large Language Models**
  Rubin Wei*, Jiaqi Cao*, **Jiarui Wang**, Jushi Kai, Qipeng Guo, Bowen Zhou, Zhouhan Lin
  ICLR 2026.
- **Memory Decoder: A Pretrained, Plug-and-Play Memory for Large Language Models**
  Jiaqi Cao*, **Jiarui Wang***, Rubin Wei, Qipeng Guo, Kai Chen, Bowen Zhou, Zhouhan Lin
  NeurIPS 2025.

*Equal contribution.

Research Experience

LLM Research Intern, Shanghai AI Lab — Aug. 2025 – Present

- Led research and development of *Intern-S2-Preview*'s memory architecture, from model training to production deployment. It supports continual expansion into new scientific domains via independently trained, plug-and-play memories, enhancing capabilities on frontier scientific tasks.
- Contributed to next-generation LLM pretraining, developing large-scale data pipelines and training models.

Research Assistant, DDST Lab, Shanghai Jiao Tong University — May. 2023 – Nov. 2023

- Developed DebugPilot, an interactive debugging system that automates error diagnosis and debugging procedure generation using program analysis and probabilistic modeling.

Honors and Awards

Second Place, RoboCup China Open, Small Size League, 2024
Third Prize, Prototype System Competition, CCF ChinaSoft, 2023
First Prize (Provincial), Senior Division, National Olympiad in Informatics in Provinces (NOIP), 2018